

- Web-based materials/chemistry simulation platform

Materials Square Class Curriculum

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Learning Objectives:

→ Confirm the thermodynamic stability of the material by calculating the cohesive energy required to separate the constituent atoms from the solid.

Materials: LiFePO_4 Bulk



MatSQ Credit: ~ 10 CPU hour (\$ 2.5)

[Lab.1] Cohesive energy

LiFePO₄-bulk

1. Load crystal structure using openDB
2. Calculate variable cell relaxation of bulk structures (primitive cell)
3. Build bulk structure with optimized lattice constant
4. Calculate total energy of bulk structure with optimized lattice constant
5. Calculate atomic energy of single atoms
6. Calculate cohesive energy

MatSQ Credit: ~ 10 CPU hour (\$ 2.5)

[Lab.1] Cohesive energy

1. Load crystal structure using openDB

Structure Builder ▲

Modeling Cell Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Search Structure from Open DB

Crystal Chemical

From Materials Project

Formula ▾ LiFePO4 🔍

mp-765634	LiFePO4	P2_1/c	56	607.547
mp-19017	LiFePO4	Pnma	28	284.498
mp-1661570	LiFePO4	P2_12_12_1	28	341.179
mp-585319	LiFePO4	Cc	112	1453.746

HISTORY

mp-19017 ★

Li: 4, Fe: 4, P: 4, O: 16
a,b,c (Å): 10.24, 5.97, 4.65
α,β,γ (°): 90.00, 90.00, 90.00

[Lab.1] Cohesive energy

2. Calculate variable cell relaxation of bulk structures (primitive cell)

Quantum Espresso

PWscf + DOS

Scripting Option : Template Atom info Update Structure Keyword information

Target property

Precision

Model type

Structure optimization

Option

Energy / Structure

High (long time)

Bulk

Ionic+cell optimization

☐ Spin polarization
☐ Van der Waals
☐ Electric field
☐ Spin Orbit Coupling
☐ DFT+U

&CONTROL

Calculation type

vc-relax

Max SCF steps

200

Information amount

high

Force threshold (Ry/bohr)

0.001

&IONS

Ion dynamics

bfgs

Upscale

100

&SYSTEM

$E_{\text{cut}}(\text{wfc})(\text{Ry})$

30

$E_{\text{cut}}(\text{rho})(\text{Ry})$

180

Occupations

smearing

Gaussian broadening (Ry)

0.002

&CELL

Cell dynamics

bfgs

Cell factor

2

Pressure threshold (Kbar)

0.2

Cell relax type

all

&ELECTRONS

Max iteration step

200

SCF must converge

True

Mixing mode

plain

Mixing beta

0.5

Convergence threshold (Ry)

0.000001

Starting wavefunction

random

KPOINTS

Sampling

Monkhorst-Pack

Grid

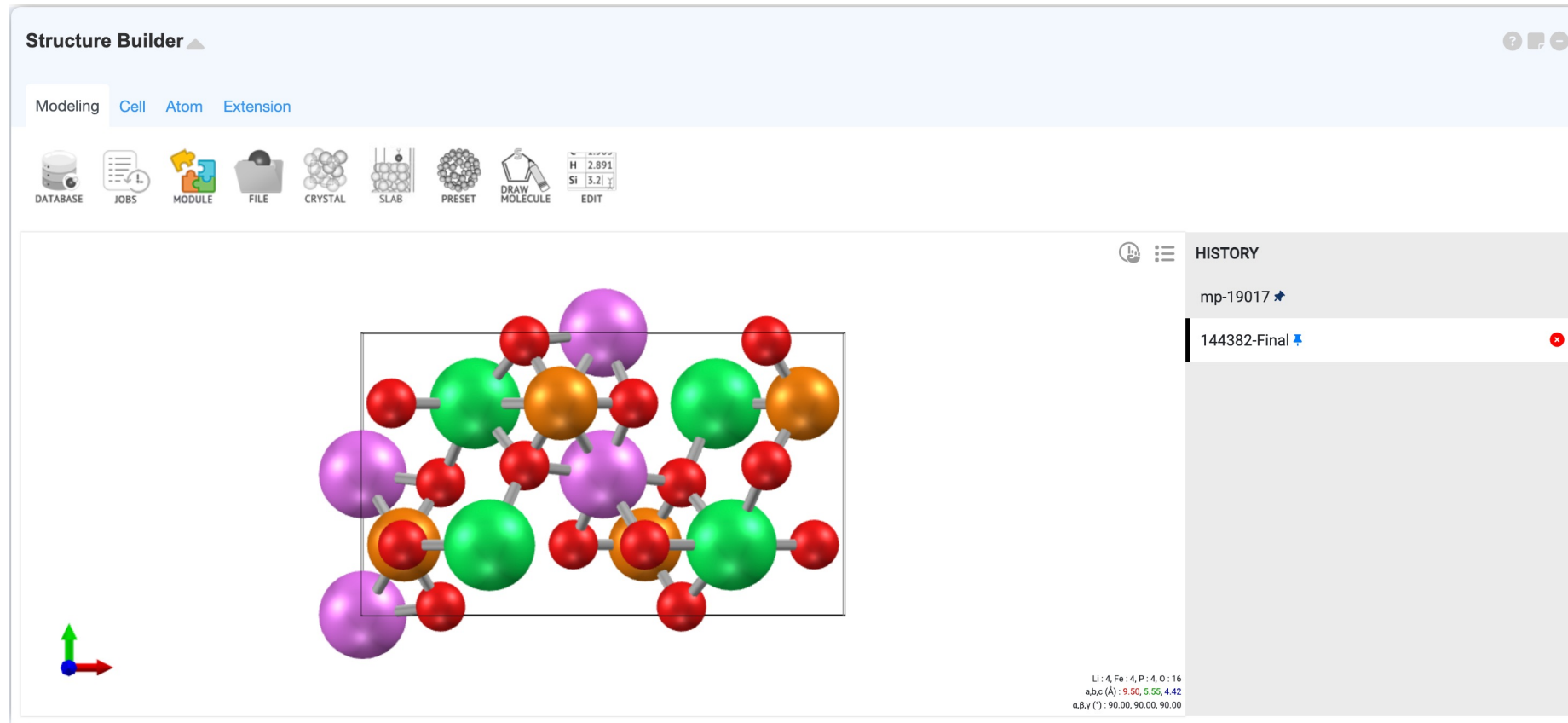
3 5 6

Shift

0 0 0

[Lab.1] Cohesive energy

3. Build bulk structure with optimized lattice constants



4. Calculate total energy of bulk structure with optimized lattice constants

Quantum Espresso

PWscf + DOS

Scripting Option : Template Atom info Update Structure Keyword information

Target property

Precision

Model type

Structure optimization

Option

Energy / Structure

High (long time)

Bulk

Ionic optimization

☒ Spin polarization
☐ Van der Waals
☐ Electric field
☐ Spin Orbit Coupling
☐ DFT+U

&CONTROL

Calculation type

Max SCF steps

Information amount

Force threshold (Ry/bohr)

relax

200

high

0.001

&SYSTEM

$E_{\text{cut}}(\text{wfc})(\text{Ry})$

$E_{\text{cut}}(\rho)(\text{Ry})$

Occupations

Gaussian broadening (Ry)

Number of electron spin

Starting magnetization

Li

Fe

P

O

30

180

smearing

0.002

2 (up,down)

1

1

1

1

&ELECTRONS

Max iteration step

SCF must converge

Mixing mode

Mixing beta

Convergence threshold (Ry)

Starting wavefunction

200

True

plain

0.5

0.000001

random

&IONS

Ion dynamics

bfgs

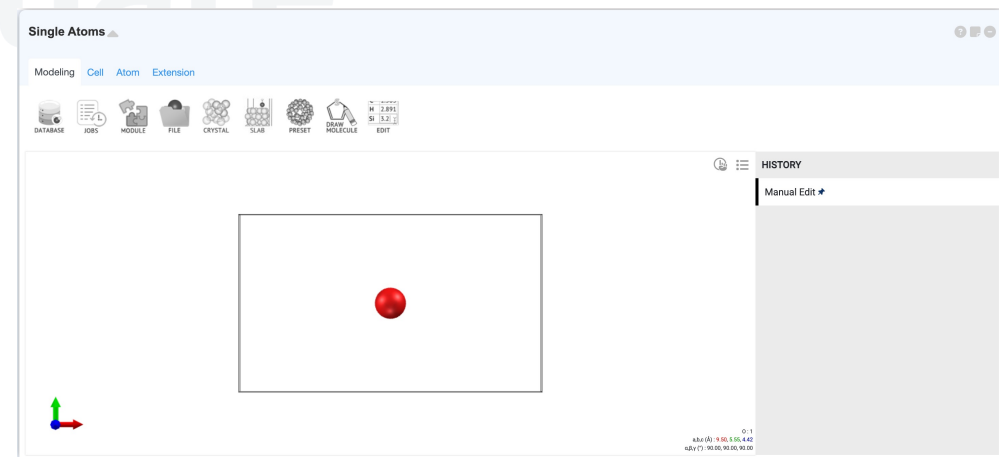
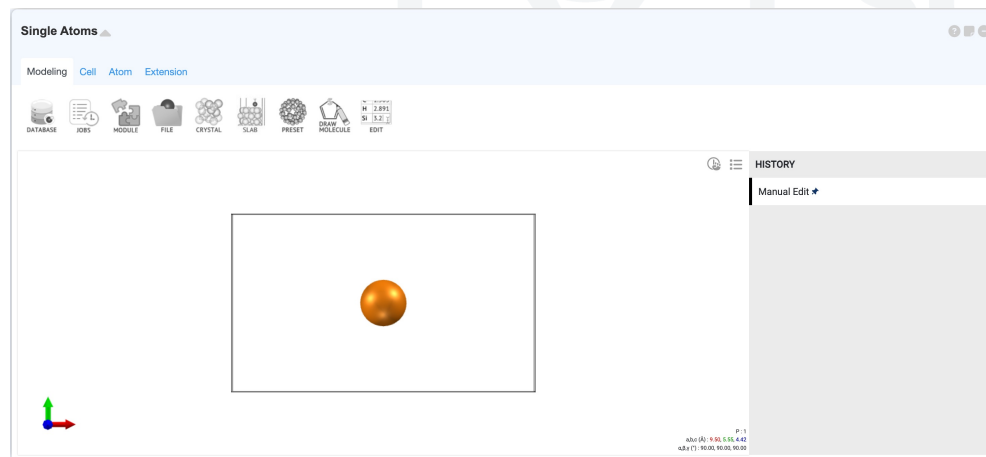
KPOINTS

Sampling

Monkhorst-Pack

[Lab.1] Cohesive energy

5. Calculate atomic energy of single atoms



5. Calculate atomic energy of single atom

High Throughput ▲

Default

PWscf

```
&CONTROL
calculation = 'scf'
forc_conv_thr = 0.00038
outdir = './output/'
prefix = 'VLAB'
pseudo_dir = './'
restart_mode = 'from_scratch'
tpnfor = .TRUE.
verbosity = 'high'
wf_collect = .TRUE.
/
```

Update

Element	Filename
Li	li_pbe_v1.4.uspp.F.UPF
Fe	Fe.pbe-spn-kjpaw_psl.0.2.1.UPF
P	P.pbe-n-rrkjus_psl.1.0.0.UPF
O	O.pbe-n-kjpaw_psl.0.1.1.UPF

Calculations

Waiting x 4

Name	Formula	Script
HT: Li-atom	Li1	pw.x
HT: Fe-atom	Fe1	pw.x
HT: P-atom	P1	pw.x
HT: O-atom	O1	pw.x

Structure

Script

Potential



6. Calculate cohesive energy

$$E_{cohesive} = \frac{1}{n_{total}} [E_{LiFePO_4} - (n_{Li} \times E_{Li\ atom} + n_{Fe} \times E_{Fe\ atom} + n_P \times E_{P\ atom} + n_O \times E_{O\ atom})] \quad n: \text{number of atoms in the bulk system}$$



[Lab.2] Li diffusion activation barrier in bulk structure

Learning Objectives:

→ Analyze the energy profile of Li diffusion between two adjacent Li sites and the corresponding Li migration path.

Materials: LiFePO₄ Bulk



MatSQ Credit: ~ 12 CPU hour (\$ 3)

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Thank you

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